

# Lorenzo Ghessi *Master's student in Automation and Control Engineering*

✉ ghesi2003@gmail.com    ☎ 347 897 8092    in lorenzoghessi    📍 Milan, Open to relocation

## ABOUT ME

MSc Automation & Control Engineering student (PoliMi + Chalmers Erasmus+), graduating April 2027. Background spans ROS/C++ software development, PLC commissioning on industrial machines, and hands-on cobot programming.

## EXPERIENCE

**Assolombarda, Laboratory Coordinator** | *Robotics Educational Lab* 09/2021 – Present | Monza

- Managed a team of 10 facilitators across shift planning and technical operations, maintaining full uptime of ABB YuMi cobots and Lab 4.0 equipment.
- Designed new didactic modules integrating ABB RobotStudio and cobot programming, expanding the curriculum adopted by Assolombarda for STEM outreach programmes.
- Delivered hands-on robotics workshops to 1000+ middle and high school students across 4+ years, achieving consistently high participant satisfaction.

**Delta Service Automation, Engineering Intern** | *PLC & HMI* 09/2024 – 10/2024 | Vaprio d'Adda

- Debugged and optimized PLC/HMI software (DELTA) on automatic labeling and product-tracking machines, directly reducing machine downtime during production runs.
- Revised electrical schematics, authored technical maintenance manuals, and modeled 3D mechanical assemblies for production line integration in SolidWorks.

## PROJECTS

**Flexible Joint Control (MATLAB/Simulink)** 02/2026 – 06/2026

Experimental parameter identification (friction, spring stiffness) and synthesis of frequency-domain and state-space controllers (LQR,  $H_\infty$ , MRAC, SMC) for precise tip positioning under uncertainty.

**Trajectory control for a Mobile Robot (ROS/C++), Milan, Italy** 10/2025 – 11/2025

Built a Turtlebot simulator in Linux/ROS Noetic (Boost Odeint); designed a PI tracking controller with feedback linearization and velocity feed-forward.

**Drone Real-Time Control (C/C++, FreeRTOS), Gothenburg, Sweden** 04/2025 – 06/2025

Developed dynamic model in Simscape, synthesized LQR controller in MATLAB/Simulink, and implemented control code on real hardware using an RTOS architecture.

**Industrial Robotic Cell 4.0 (PLC/HMI, Industrial Robotics), Sesto San Giovanni, Italy** 03/2021 – 06/2021

Designed a pick-and-place cell integrating ABB RobotStudio, Siemens S7-1200 PLC, HMI, and a computer vision system for quality inspection.

## EDUCATION

**Polytechnic University of Milan, MSc Automation and control engineering** 09/2024 – Present | Milan, Italy

**Chalmers University of Technology, Erasmus** 01/2025 – 06/2025 | Gothenburg, Sweden

**Polytechnic University of Milan, BSc in Automation and control engineering** 09/2021 – 07/2024 | Milan, Italy

## SKILLS AND CERTIFICATIONS

### Software

ROS/ROS2, MATLAB/Simulink, ABB RobotStudio, SolidWorks, Linux (Ubuntu), AutoCAD

### Programming

C/C++, Python, PLC (Ladder, ST, SFC)

### Core competencies

Multivariable control, Robot programming (ABB RAPID), Industrial commissioning, Computer Vision, Real-time systems (FreeRTOS), Teleoperation

### Languages

Italian (native), English (C1 — IELTS 7.0), Spanish (elementary), German (beginner, A1)